| Ex. No. 8 | Modern User Interface Design using ReactJS |
|---|---|
| **Date of Exercise** | 18.09.2026 |
| **GitHub Host Link** | https://jinto-joseph.github.io/WEB_TECH_LAB/Exp_08/react-ui-lab/index.html |

## Aim

To design and develop a modern, responsive user interface using ReactJS by applying the following concepts:

- React components
- JSX
- Props
- State
- Event handling
- CSS styling
- Responsive design
- Reusable UI components

## 1. Software Required

- Node.js
- Visual Studio Code
- Web browser such as Chrome or Edge

Verify Node.js and npm installation using:

```
node -v
```

```
npm -v
```

## 2. Create a React Project

For a modern React setup, use Vite. Open the terminal and run:

```
npm create vite@latest react-ui-lab
```

Choose:

```
Framework: React
```

```
Variant: JavaScript
```

Move into the project folder, install dependencies, and start the development server:

cd react-ui-lab

npm install

npm run dev

The terminal will display a local URL, typically similar to:

http://localhost:5173/

Open this URL in a web browser.

## 3. Explain the Project Structure

```
react-ui-lab
│
├── src
│   ├── App.jsx
│   ├── App.css
│   ├── index.css
│   └── main.jsx
│
├── public
├── package.json
└── vite.config.js
```

| File | Purpose |
|---|---|
| main.jsx | Entry point of the React application |
| App.jsx | Main application component |
| App.css | Styling for the App component |
| index.css | Global CSS styles |
| package.json | Project dependencies and scripts |

## 4. Clean the Default React Code

Open src/App.jsx and replace the existing content with:

```
function App() {
  return (
    <div>
      <h1>Student Dashboard</h1>
    </div>
  );
}

export default App;
```

Run the project and confirm that the browser displays “Student Dashboard”.

## 5. Explain JSX

JSX allows HTML-like syntax to be written inside JavaScript. Example:

```
const studentName = "Arun";

function App() {
  return (
    <div>
      <h1>Welcome, {studentName}</h1>
    </div>
  );
}
```

The expression {studentName} inserts a JavaScript value into JSX.

## 6. Create a Modern Dashboard Layout

Replace App.jsx with the following dashboard structure:

```
import "./App.css";

function App() {
  return (
    <div className="app">
      <header className="header">
        <h2>EduDashboard</h2>
        <button className="profile-btn">Student Profile</button>
      </header>

      <main className="dashboard">
        <section className="welcome">
          <h1>Welcome Back!</h1>
          <p>Here is your academic overview.</p>
        </section>

        <section className="cards">
          <div className="card"><h3>Courses</h3><p>6</p></div>
          <div className="card"><h3>Assignments</h3><p>12</p></div>
          <div className="card"><h3>Attendance</h3><p>92%</p></div>
          <div className="card"><h3>CGPA</h3><p>8.7</p></div>
        </section>
      </main>
    </div>
  );
}

export default App;
```

## 7. Add Modern CSS Styling

Open src/App.css and use the following styles:

```
* {
  margin: 0;
  padding: 0;
  box-sizing: border-box;
}

body {
  font-family: Arial, sans-serif;
  background: #f4f7fb;
}

.app { min-height: 100vh; }

.header {
  display: flex;
  justify-content: space-between;
  align-items: center;
  padding: 20px 40px;
  background: white;
  box-shadow: 0 2px 8px rgba(0, 0, 0, 0.08);
}

.header h2 { color: #333; }

.profile-btn {
  border: none;
  background: #4f46e5;
  color: white;
  padding: 10px 18px;
  border-radius: 8px;
  cursor: pointer;
}

.profile-btn:hover { background: #3730a3; }

.dashboard { padding: 40px; }
.welcome { margin-bottom: 30px; }
.welcome h1 { font-size: 32px; margin-bottom: 8px; }
.welcome p { color: #666; }

.cards {
  display: grid;
  grid-template-columns: repeat(4, 1fr);
  gap: 20px;
}

.card {
  background: white;
  padding: 25px;
  border-radius: 14px;
  box-shadow: 0 4px 15px rgba(0, 0, 0, 0.06);
  transition: transform 0.3s ease;
}

.card:hover { transform: translateY(-5px); }
.card h3 { color: #555; margin-bottom: 15px; }
.card p { font-size: 28px; font-weight: bold; color: #4f46e5; }
```

At this stage, students should see a modern card-based dashboard.

## 8. Teach Reusable Components

Create a folder named src/components and create StatCard.jsx inside it:

```
function StatCard({ title, value }) {
  return (
    <div className="card">
      <h3>{title}</h3>
      <p>{value}</p>
    </div>
  );
}

export default StatCard;
```

Modify App.jsx to use the reusable component:

```
import "./App.css";
import StatCard from "./components/StatCard";

function App() {
  return (
    <div className="app">
      <header className="header">
        <h2>EduDashboard</h2>
        <button className="profile-btn">Student Profile</button>
      </header>

      <main className="dashboard">
        <section className="welcome">
          <h1>Welcome Back!</h1>
          <p>Here is your academic overview.</p>
        </section>

        <section className="cards">
          <StatCard title="Courses" value="6" />
          <StatCard title="Assignments" value="12" />
          <StatCard title="Attendance" value="92%" />
          <StatCard title="CGPA" value="8.7" />
        </section>
      </main>
    </div>
  );
}

export default App;
```

## 9. Explain Props

The component call below passes data from the parent to the child component:

<StatCard title = “Course” value = “6” />

The child receives those values using:

function StatCard({ title, value })

Props are data passed from a parent component to a child component.

## 10. Add State

Import useState and add theme switching:

```
import { useState } from "react";
import "./App.css";
import StatCard from "./components/StatCard";

function App() {
  const [darkMode, setDarkMode] = useState(false);

  return (
    <div className={darkMode ? "app dark" : "app"}>
      <header className="header">
        <h2>EduDashboard</h2>

        <button
          className="profile-btn"
          onClick={() => setDarkMode(!darkMode)}
        >
          Toggle Theme
        </button>
      </header>

      <main className="dashboard">
        <section className="welcome">
          <h1>Welcome Back!</h1>
          <p>Here is your academic overview.</p>
        </section>

        <section className="cards">
          <StatCard title="Courses" value="6" />
          <StatCard title="Assignments" value="12" />
          <StatCard title="Attendance" value="92%" />
          <StatCard title="CGPA" value="8.7" />
        </section>
      </main>
    </div>
  );
}

export default App;
```

## 11. Add Dark Mode CSS

```
.dark {
  background: #111827;
  color: white;
}

.dark .header { background: #1f2937; }
.dark .header h2 { color: white; }
.dark .card { background: #1f2937; }
.dark .card h3 { color: #d1d5db; }
.dark .welcome p { color: #d1d5db; }
```

Clicking the button should switch between light and dark themes.

## 12. Explain State

const [darkMode, setDarkMode] = useState(false);

| Part | Meaning |
|---|---|
| darkMode | Current state value |
| setDarkMode | Function used to update the value |
| false | Initial state value |

When setDarkMode(!darkMode) executes, React updates the state and re-renders the interface.

## 13. Add a Student Course List

Add the following array inside App.jsx:

```
const courses = [
  "Web Technology",
  "Database Management",
  "Machine Learning",
  "Computer Networks"
];
```

Then render the courses inside the dashboard:

```
<section className="course-section">
  <h2>My Courses</h2>

  <div className="course-list">
    {courses.map((course, index) => (
      <div className="course-item" key={index}>
        {course}
      </div>
    ))}
  </div>
</section>
```

## 14. Style the Course List

```
.course-section { margin-top: 40px; }
.course-section h2 { margin-bottom: 20px; }

.course-list {
  display: grid;
  grid-template-columns: repeat(2, 1fr);
  gap: 15px;
}

.course-item {
  background: white;
  padding: 18px;
  border-radius: 10px;
  box-shadow: 0 3px 10px rgba(0, 0, 0, 0.05);
}

.dark .course-item { background: #1f2937; }
```

This introduces list rendering using map().

## 15. Add an Input Form

Add the following state variable:

const [task, setTask] = useState(“”);

Then add the form section:

<section className = “task-section”>

<h2>Add Task</h2>

<input

Type = “text”

Placeholder = “Enter a task :”

value={task}

onChange={(event) => setTask(event.target.value)}

/>

<button onClick={() => alert(task)}>Add</button>

</section>

## 16. Style the Form

```
.task-section { margin-top: 40px; }
.task-section h2 { margin-bottom: 15px; }

.task-section input {
  padding: 12px;
  width: 280px;
  border: 1px solid #ccc;
  border-radius: 8px;
  margin-right: 10px;
}

.task-section button {
  padding: 12px 20px;
  border: none;
  border-radius: 8px;
  background: #4f46e5;
  color: white;
  cursor: pointer;
}
```

## 17. Teach Controlled Components

The input value is controlled by React state:

value={task}

onChange={(event) => setTask(event.target.value)}

This is called a controlled component because React state controls the input value.

## 18. Make the Interface Responsive

Add media queries so the interface adapts to tablet and mobile screens:

```
@media (max-width: 900px) {
  .cards {
    grid-template-columns: repeat(2, 1fr);
  }
}

@media (max-width: 600px) {
  .header { padding: 15px 20px; }
  .dashboard { padding: 20px; }
```

```
  .cards { grid-template-columns: 1fr; }
  .course-list { grid-template-columns: 1fr; }

  .task-section input {
    width: 100%;
    margin-bottom: 10px;
  }
}
```

Resize the browser window to verify that the layout remains usable at different screen sizes.

## 19. Explain Modern UI Principles

| Principle | Meaning |
|---|---|
| Consistency | Similar buttons, cards, and controls should use consistent styling. |
| Spacing | Use adequate whitespace and avoid overcrowded layouts. |
| Contrast | Text and controls should be easy to distinguish and read. |
| Typography | Use a clear hierarchy of font sizes and weights. |
| Responsiveness | The interface should adapt to different screen sizes. |
| Feedback | Controls should provide visible feedback to user actions. |
| Reusability | Repeated interface patterns should be implemented as reusable components. |
| Accessibility | The interface should be usable by people with different abilities. |

React provides component structure and interactivity; a modern user interface comes from sound layout, spacing, typography, responsiveness, accessibility, and interaction design.

## 20. Add Hover Effects

```
.profile-btn,
.task-section button {
  transition: all 0.3s ease;
}

.profile-btn:hover,
.task-section button:hover {
  transform: translateY(-2px);
  opacity: 0.9;
}
```

## 21. Add Navigation

Create Navbar.jsx inside src/components:

```
function Navbar() {
  return (
    <nav className="navbar">
      <h2>EduDashboard</h2>

      <div className="nav-links">
        <a href="#">Home</a>
        <a href="#">Courses</a>
        <a href="#">Tasks</a>
        <a href="#">Profile</a>
      </div>
    </nav>
  );
}

export default Navbar;
```

Add the following CSS:

```css
.navbar {
  display: flex;
  align-items: center;
  justify-content: space-between;
}

.nav-links {
  display: flex;
  gap: 20px;
}

.nav-links a {
  text-decoration: none;
  color: #444;
}
```

## 22. Final Suggested Project Structure

```
src
│
├── components
│   ├── Navbar.jsx
│   └── StatCard.jsx
│
├── App.jsx
├── App.css
├── index.css
└── main.jsx
```

## 23. Student Lab Activity

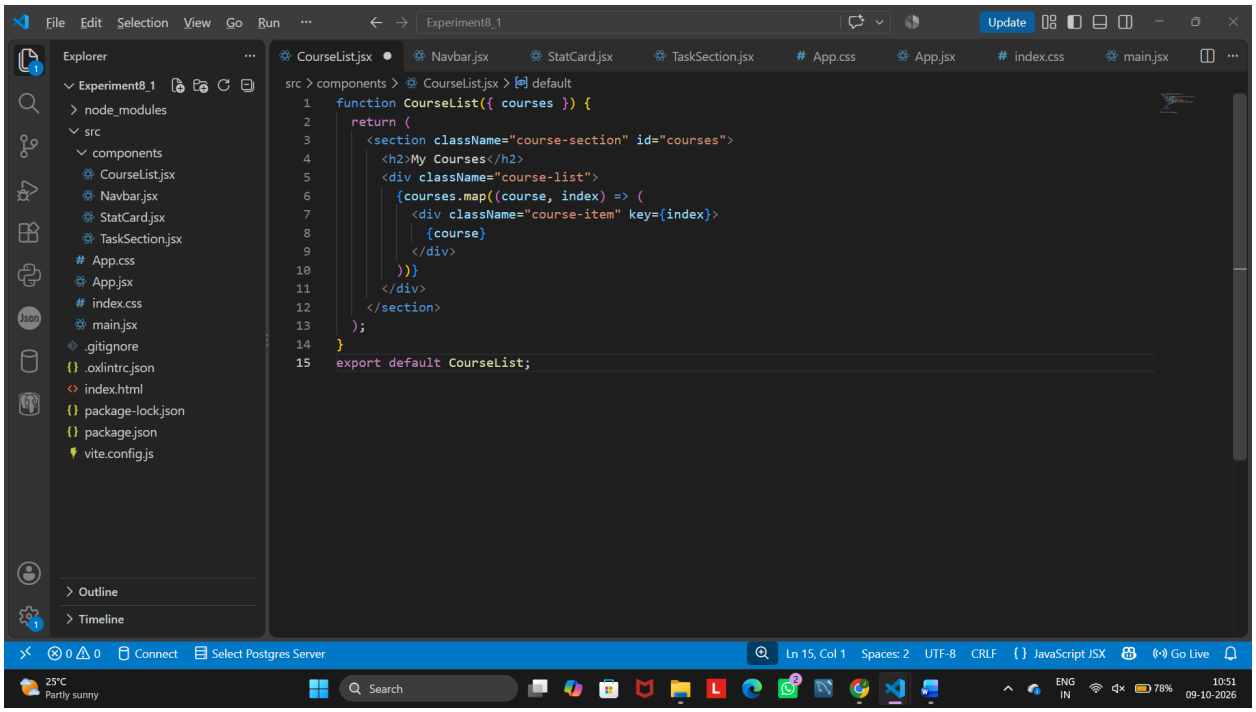
Problem: Design a modern Student Dashboard using ReactJS. The application should contain:

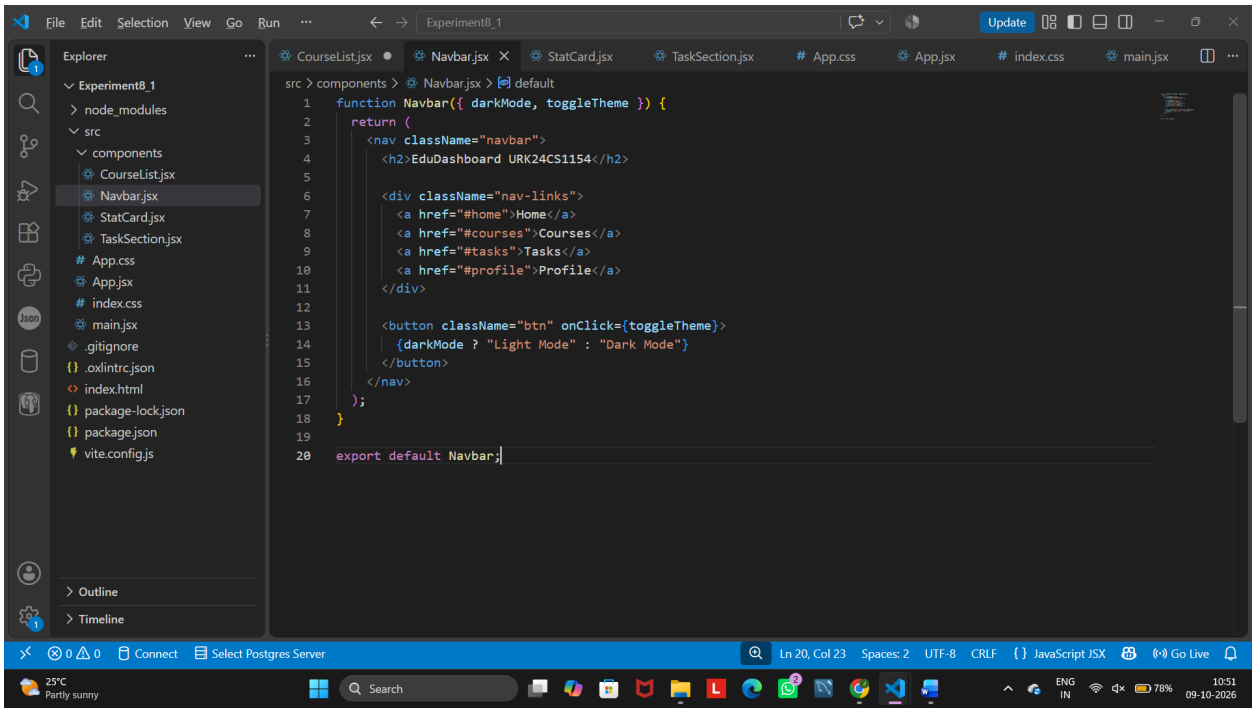
1. Navigation bar
2. Welcome section
3. At least four statistic cards
4. Course list
5. Task input field
6. Button event
7. Light/dark theme
8. Responsive layout

9. Hover effects

10. Reusable components

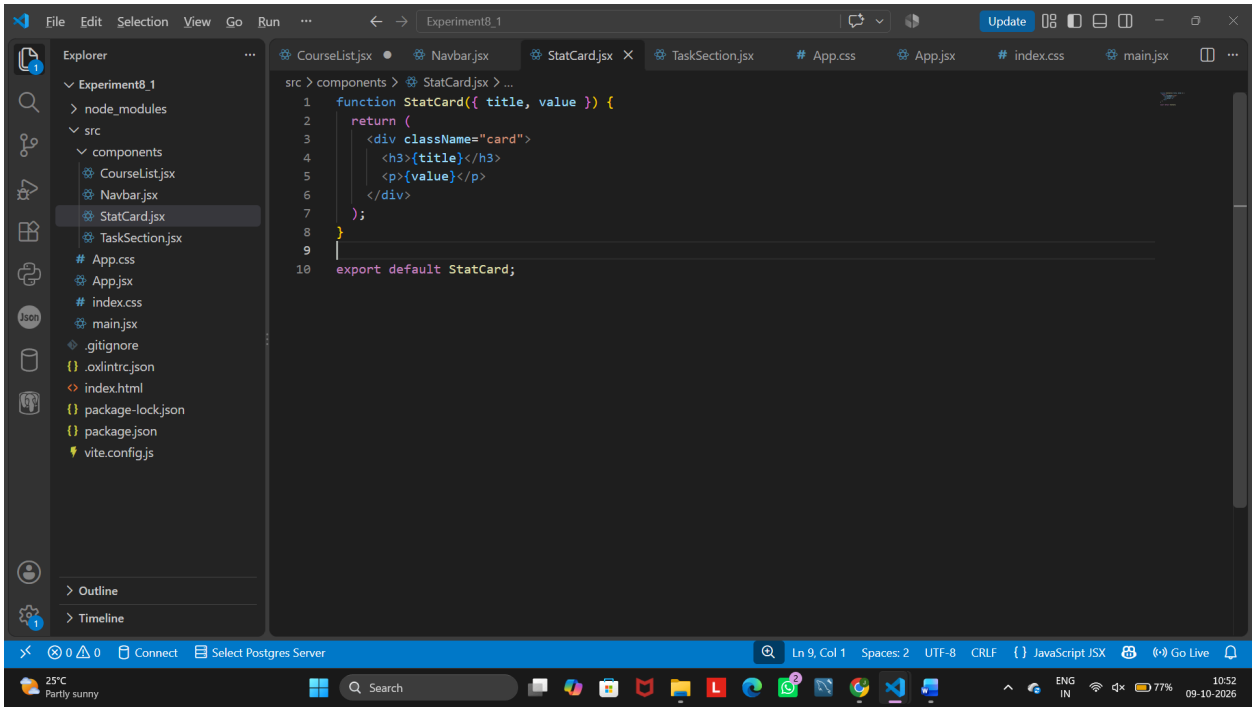
## Program




File Edit Selection View Go Run
Experiment8_1
Update
Explorer
Experiment8_1
node_modules
src
components
CourseList.jsx
Navbar.jsx
StatCard.jsx
TaskSection.jsx
App.css
App.jsx
index.css
main.jsx
.gitignore
.oxlintrc.json
index.html
package-lock.json
package.json
vite.config.js
CourseList.jsx
Navbar.jsx
StatCard.jsx
TaskSection.jsx
App.css
App.jsx
index.css
main.jsx
src > components > Navbar.jsx > default
function Navbar({ darkMode, toggleTheme }) {
return (
<nav className="navbar">
<h2>EduDashboard URK24CS1154</h2>

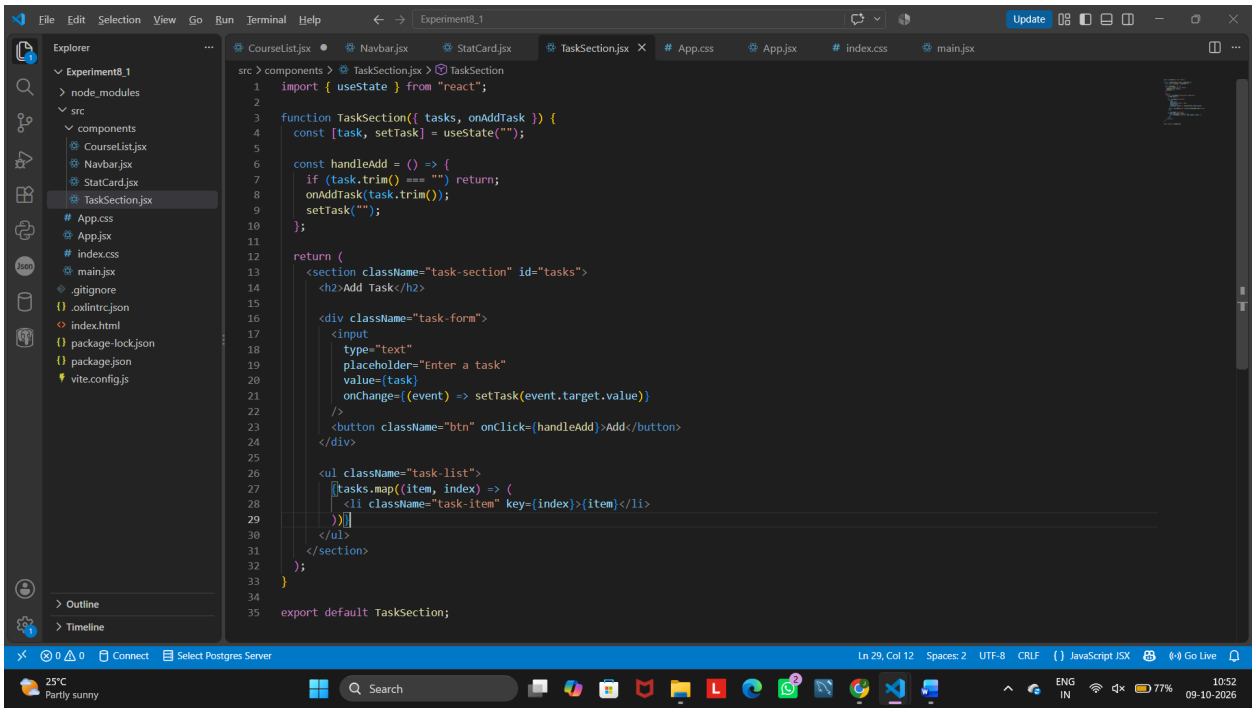
<a href="#home">Home</a>
<a href="#courses">Courses</a>
<a href="#tasks">Tasks</a>
<a href="#profile">Profile</a>

<button className="btn" onClick={toggleTheme}>
{darkMode ? "Light Mode" : "Dark Mode"}
</button>
</nav>
);
}
export default Navbar;
Outline
Timeline
0 0 Connect Select Postgres Server
Ln 20, Col 23 Spaces: 2 UTF-8 CRLF JavaScript JSX Go Live
25°C Partly sunny
Search
ENG IN
78%
10:51
09-10-2026


File Edit Selection View Go Run
Experiment8_1
Update
Explorer
Experiment8_1
node_modules
src
components
CourseList.jsx
Navbar.jsx
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TaskSection.jsx
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App.jsx
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package.json
vite.config.js
CourseList.jsx
Navbar.jsx
StatCard.jsx
TaskSection.jsx
App.css
App.jsx
index.css
main.jsx
src > components > StatCard.jsx > ...
function StatCard({ title, value }) {
return (

<h3>{title}</h3>
{value}

);
}
export default StatCard;
Outline
Timeline
0 0 Connect Select Postgres Server
Ln 9, Col 1 Spaces: 2 UTF-8 CRLF JavaScript JSX Go Live
25°C Partly sunny
Search
ENG IN
77%
10:52
09-10-2026



```jsx
import { useState } from "react";

function TaskSection({ tasks, onAddTask }) {
  const [task, setTask] = useState("");

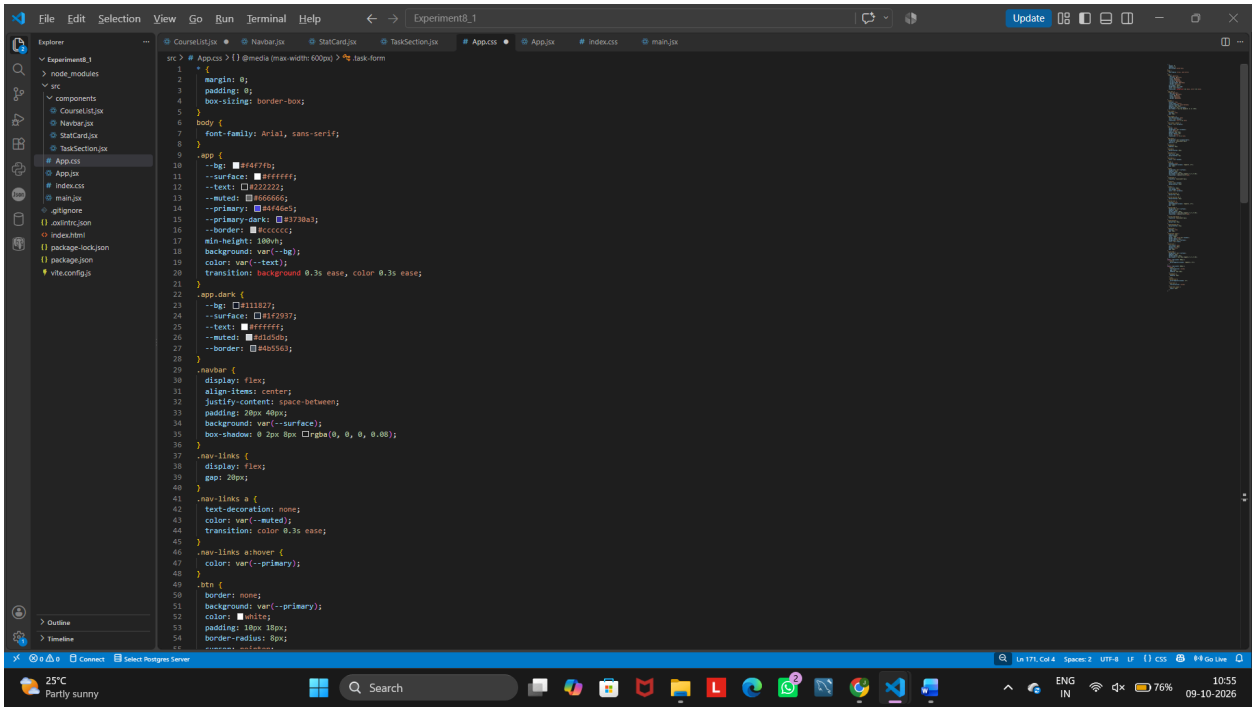
  const handleAdd = () => {
    if (task.trim() === "") return;
    onAddTask(task.trim());
    setTask("");
  };

  return (
    <section className="task-section" id="tasks">
      <h2>Add Task</h2>

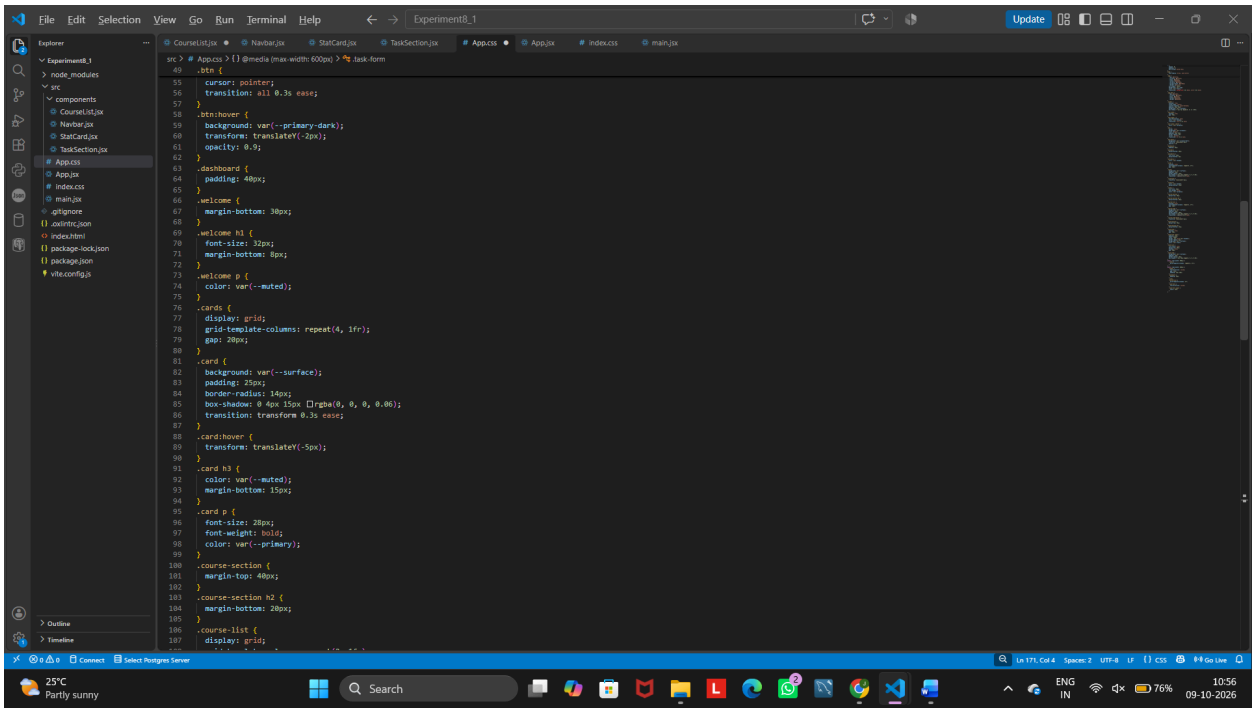
      <div className="task-form">
        <input
          type="text"
          placeholder="Enter a task"
          value={task}
          onChange={(event) => setTask(event.target.value)}
        />
        <button className="btn" onClick={handleAdd}>Add</button>
      </div>

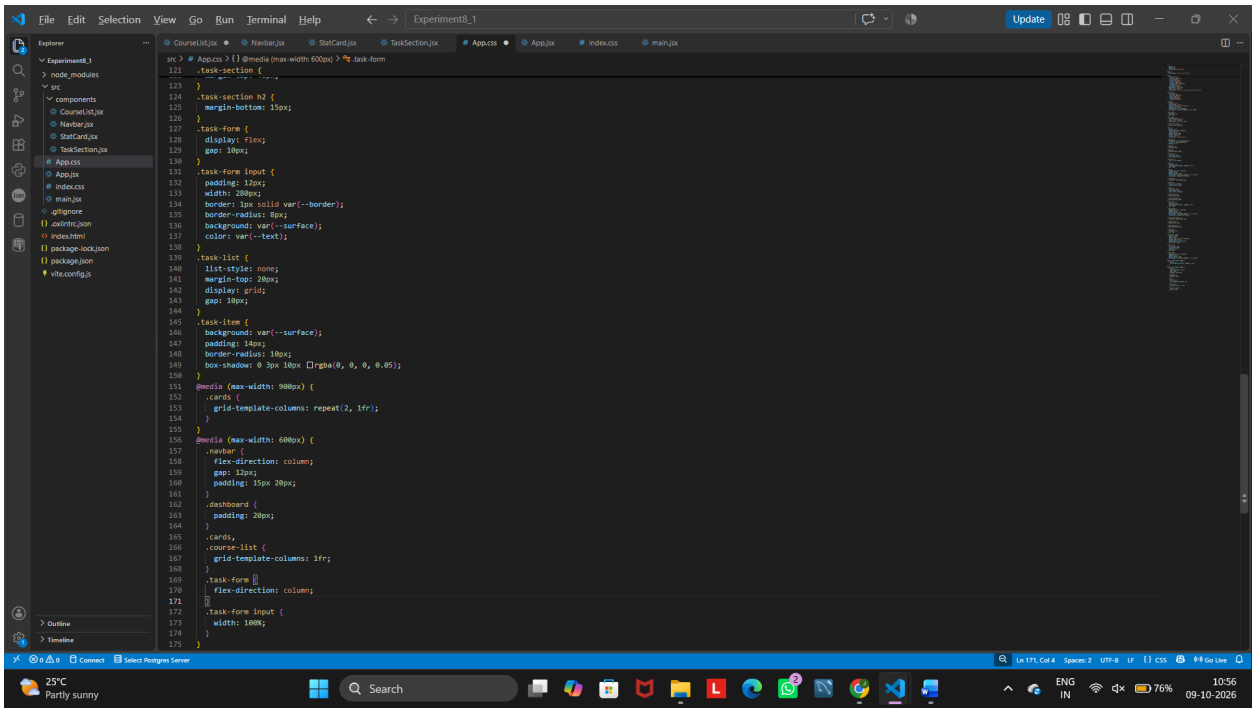
      <ul className="task-list">
        {tasks.map((item, index) => (
          <li className="task-item" key={index}>{item}</li>
        ))}
      </ul>
    </section>
  );
}

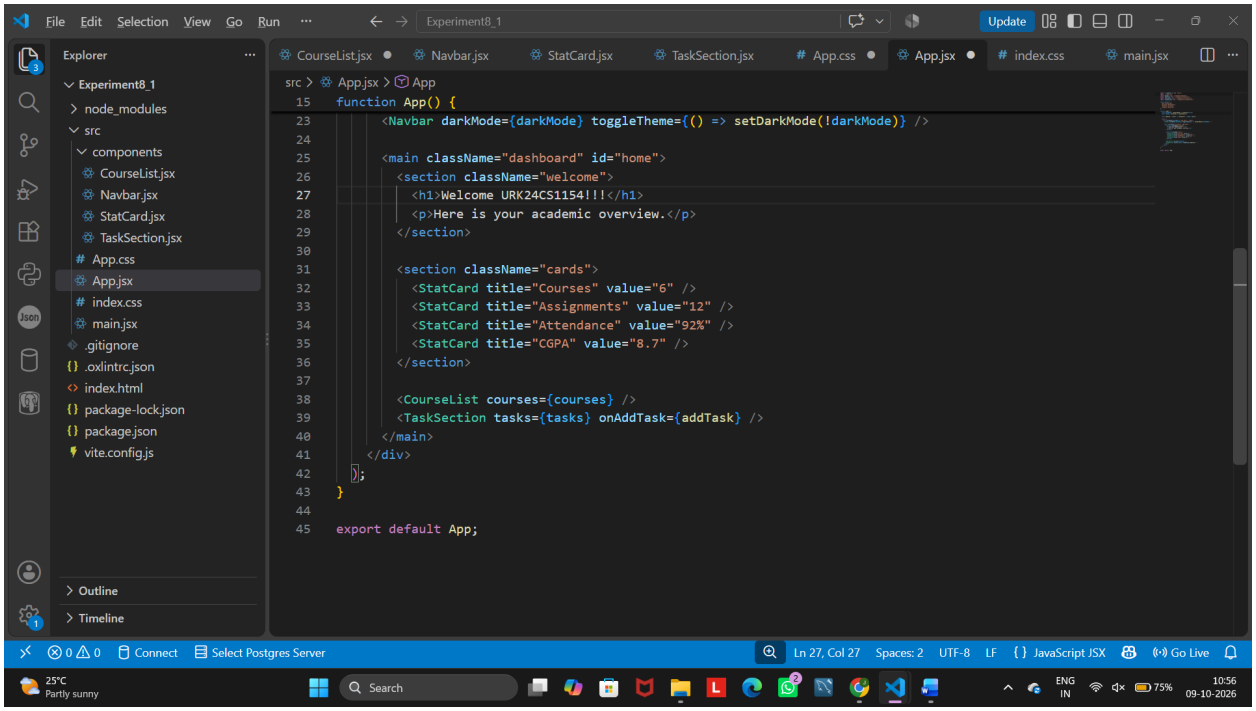
export default TaskSection;
```



```css
* {
  margin: 0;
  padding: 0;
  box-sizing: border-box;
}
body {
  font-family: Arial, sans-serif;
}
.app {
  --bg: #f4f7fb;
  --surface: #ffffff;
  --text: #222222;
  --muted: #666666;
  --primary: #4f46e5;
  --primary-dark: #3730a3;
  --border: #cccccc;
  min-height: 100vh;
  background: var(--bg);
  color: var(--text);
  transition: background 0.3s ease, color 0.3s ease;
}
.app.dark {
  --bg: #111827;
  --surface: #1f2937;
  --text: #ffffff;
  --muted: #d1d5db;
  --border: #4b5563;
}
.navbar {
  display: flex;
  align-items: center;
  justify-content: space-between;
  padding: 20px 40px;
  background: var(--surface);
  box-shadow: 0 2px 8px rgba(0, 0, 0, 0.08);
}
.nav-links {
  display: flex;
  gap: 20px;
}
.nav-links a {
  text-decoration: none;
  color: var(--muted);
  transition: color 0.3s ease;
}
.nav-links a:hover {
  color: var(--primary);
}
.btn {
  border: none;
  background: var(--primary);
  color: white;
  padding: 10px 18px;
  border-radius: 8px;
```

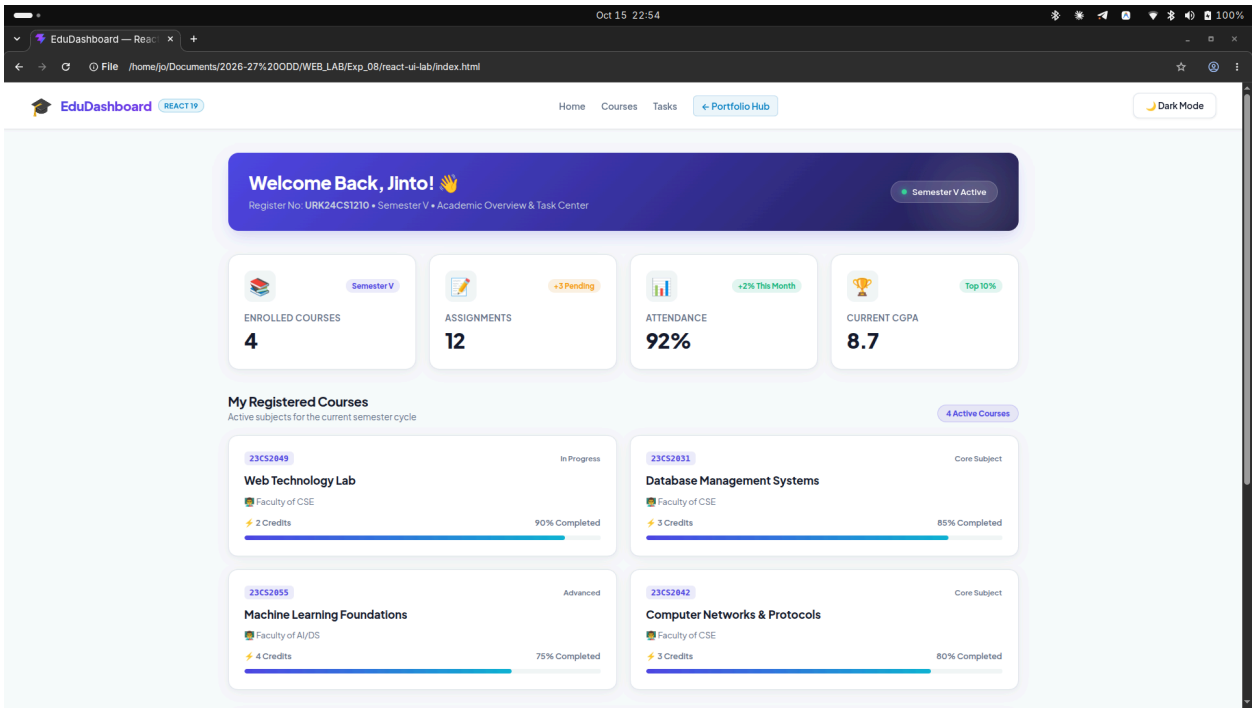

File Edit Selection View Go Run Terminal Help
Update
Explorer
Experiment8_1
node_modules
src
components
CourseList.jsx
Navbar.jsx
StatCard.jsx
TaskSection.jsx
App.css
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index.css
main.jsx
.gitignore
.oxlintrc.json
index.html
package-lock.json
package.json
vite.config.js
Outline
Timeline
.btn {
cursor: pointer;
transition: all 0.3s ease;
}
.btn:hover {
background: var(--primary-dark);
transform: translateY(-2px);
opacity: 0.9;
}
.dashboard {
padding: 40px;
}
.welcome {
margin-bottom: 30px;
}
.welcome h1 {
font-size: 32px;
margin-bottom: 8px;
}
.welcome p {
color: var(--muted);
}
.cards {
display: grid;
grid-template-columns: repeat(4, 1fr);
gap: 20px;
}
.card {
background: var(--surface);
padding: 25px;
border-radius: 14px;
box-shadow: 0 4px 15px rgba(0, 0, 0, 0.06);
transition: transform 0.3s ease;
}
.card:hover {
transform: translateY(-5px);
}
.card h3 {
color: var(--muted);
margin-bottom: 15px;
}
.card p {
font-size: 28px;
font-weight: bold;
color: var(--primary);
}
.course-section {
margin-top: 40px;
}
.course-section h2 {
margin-bottom: 20px;
}
.course-list {
display: grid;
25°C
Partly sunny
Search
ENG
IN
76%
10:56
09-10-2026

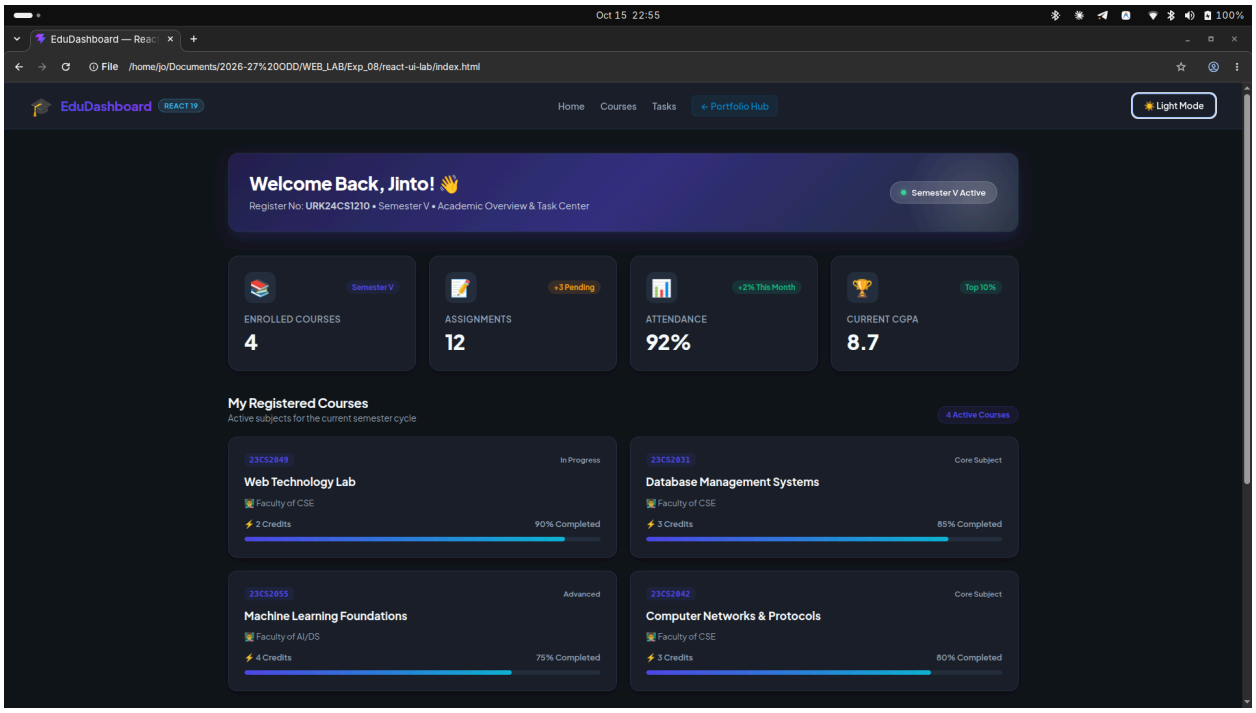



File Edit Selection View Go Run
Update
Explorer
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node_modules
src
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TaskSection.jsx
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index.html
package-lock.json
package.json
vite.config.js
Outline
Timeline
CourseList.jsx Navbar.jsx StatCard.jsx TaskSection.jsx App.css App.jsx index.css main.jsx
src > App.jsx > App
function App() {
<Navbar darkMode={darkMode} toggleTheme={() => setDarkMode(!darkMode)} />
<main className="dashboard" id="home">
<section className="welcome">
<h1>Welcome URK24CS1154!!!</h1>
Here is your academic overview.
</section>
<section className="cards">
<StatCard title="Courses" value="6" />
<StatCard title="Assignments" value="12" />
<StatCard title="Attendance" value="92%" />
<StatCard title="CGPA" value="8.7" />
</section>
<CourseList courses={courses} />
<TaskSection tasks={tasks} onAddTask={addTask} />
</main>

);
}
export default App;
Ln 27, Col 27 Spaces: 2 UTF-8 LF JavaScript JSX Go Live
Connect
Select Postgres Server
25°C Partly sunny
Search
ENG IN
10:56 09-10-2026

## Output



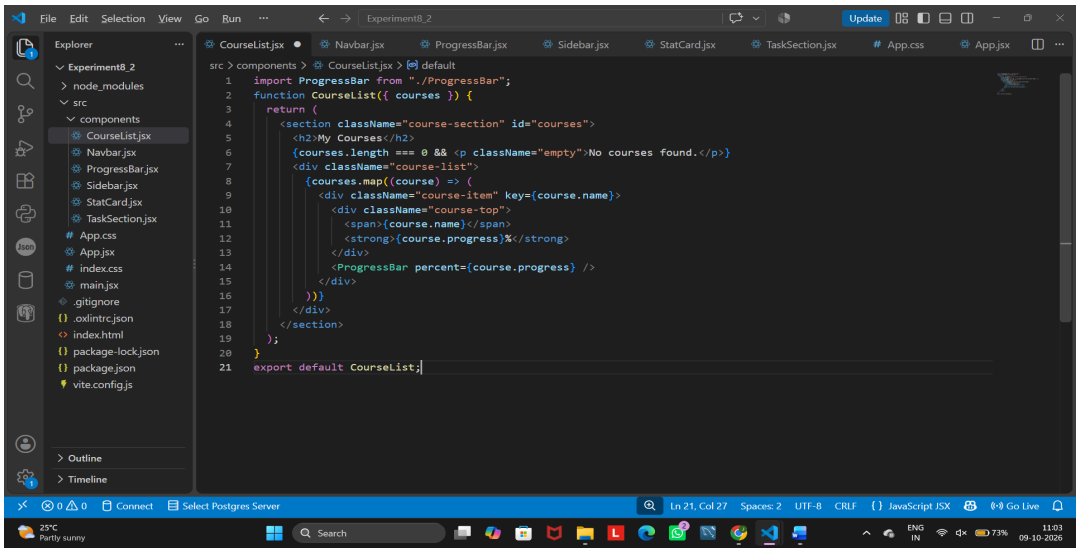



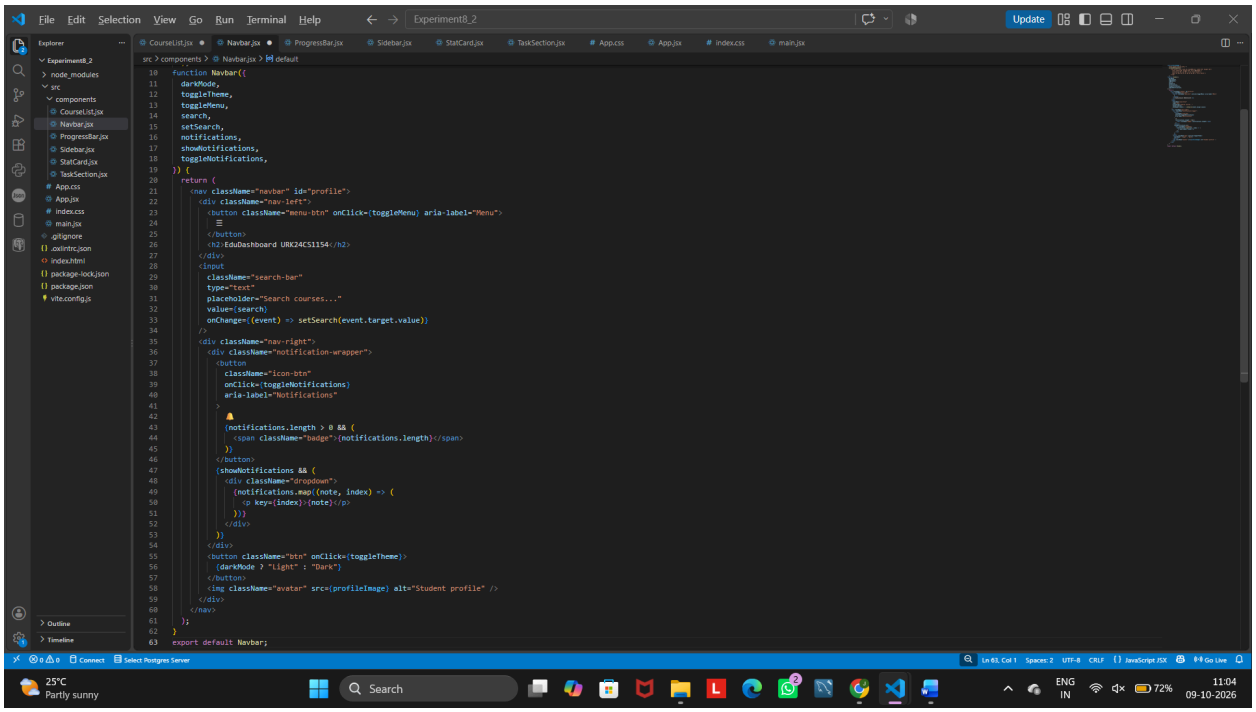

## 24. Additional Student Challenge

Ask students to improve the application by adding the following features:

- Profile image
- Search bar
- Notification icon
- Progress bars
- Course completion percentage
- Task deletion
- Task completion
- Sidebar
- Responsive menu

## Program




File Edit Selection View Go Run Terminal Help
Experiment8_2
Update
Explorer
Experiment8_2
node_modules
src
components
CourseList.jsx
Navbar.jsx
ProgressBar.jsx
Sidebar.jsx
StatCard.jsx
TaskSection.jsx
App.css
App.jsx
index.css
main.jsx
.gitignore
.oxlintrc.json
index.html
package-lock.json
package.json
vite.config.js
Outline
Timeline
function Navbar({
darkMode,
toggleTheme,
toggleMenu,
search,
setSearch,
notifications,
showNotifications,
toggleNotifications,
}) {
return (
<nav className="navbar" id="profile">

<button className="menu-btn" onClick={toggleMenu} aria-label="Menu">
☰
</button>
<h2>EduDashboard URK24CS1154</h2>

<input
className="search-bar"
type="text"
placeholder="Search courses..."
value={search}
onChange={(event) => setSearch(event.target.value)}
/>

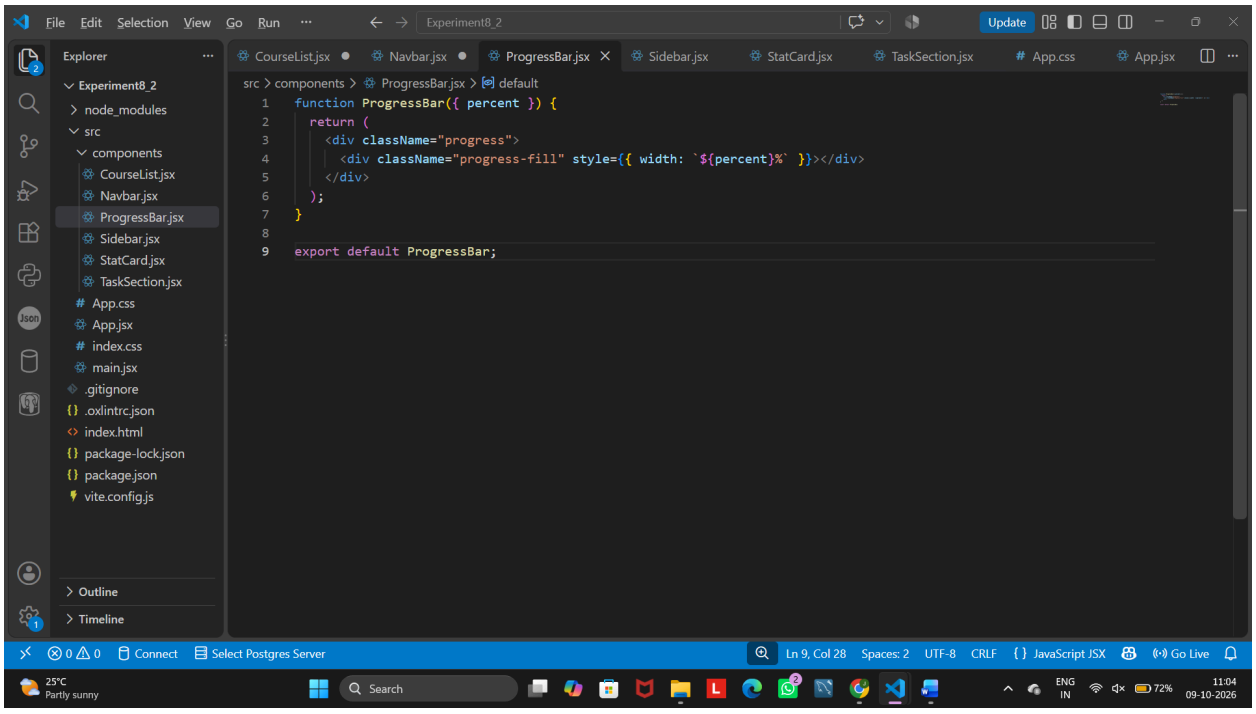
<button
className="icon-btn"
onClick={toggleNotifications}
aria-label="Notifications"
>
🔔
{notifications.length > 0 && (
{notifications.length}
)}
</button>
{showNotifications && (

{notifications.map((note, index) => (
{note}
))}

)}

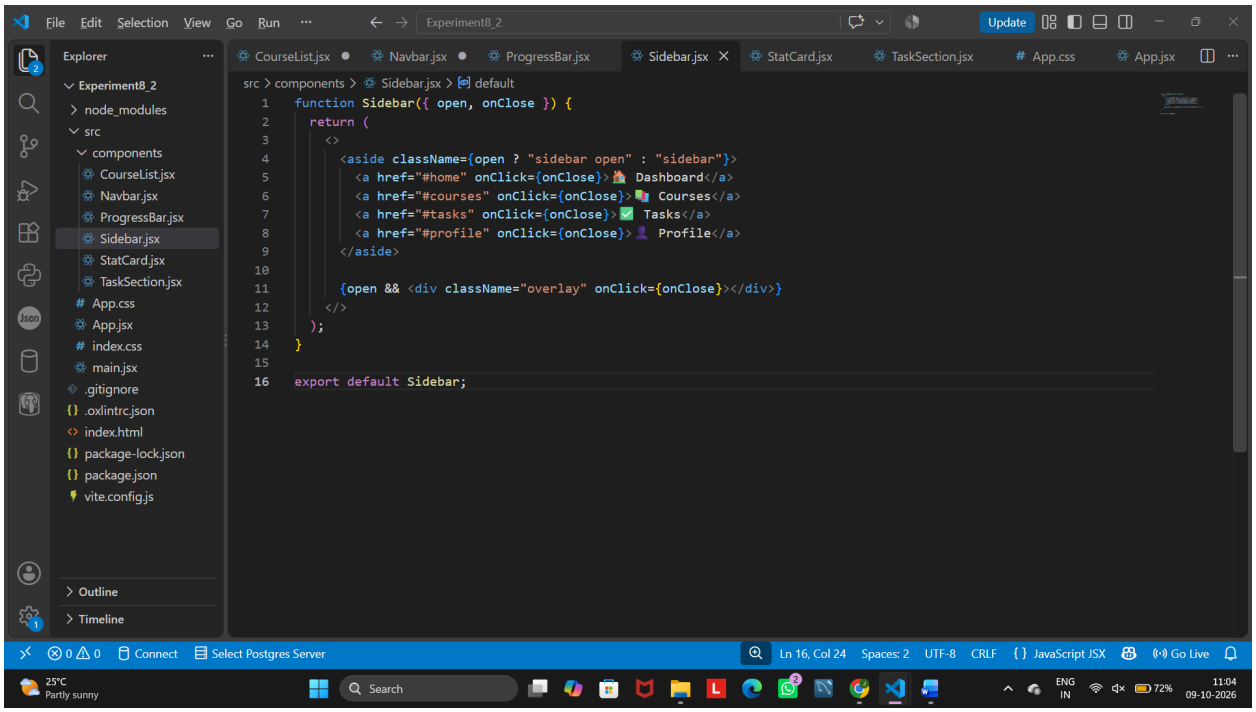
<button className="btn" onClick={toggleTheme}>
{darkMode ? "Light" : "Dark"}
</button>
<img className="avatar" src={profileImage} alt="Student profile" />

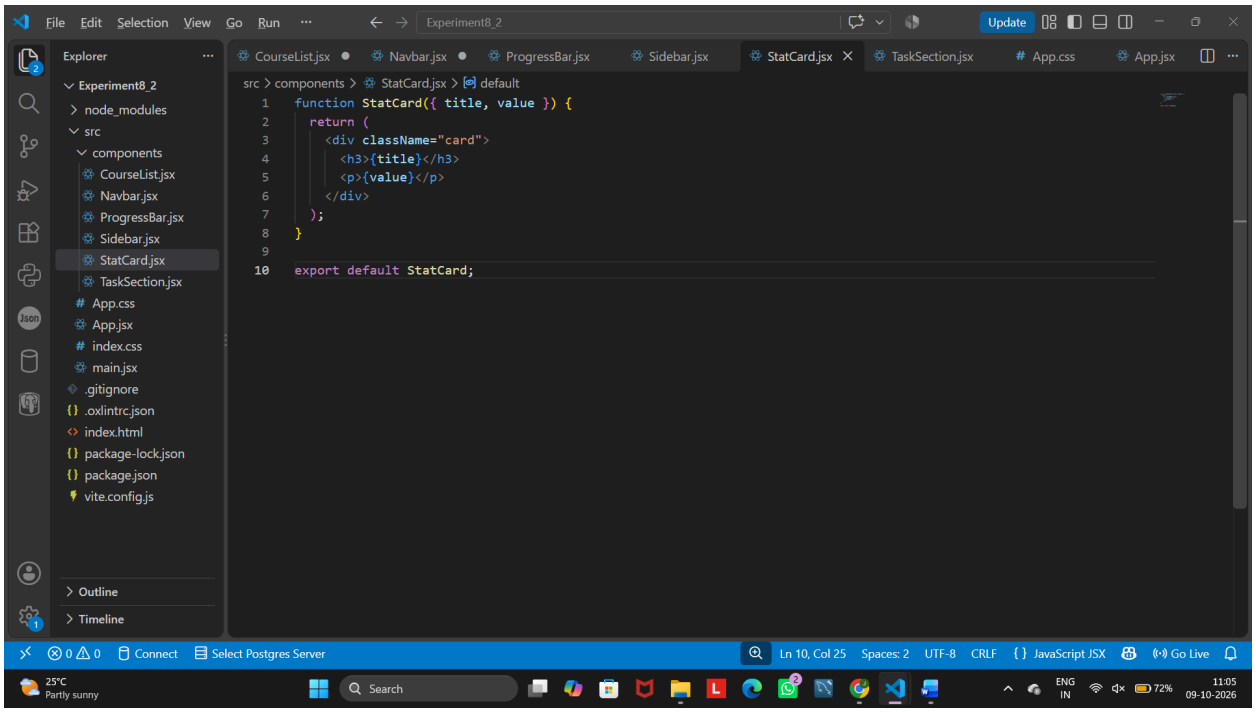
</nav>
);
}
export default Navbar;
Ln 63, Col 1
Spaces: 2
UTF-8
CRLF
JavaScript JSX
Go Live
25°C
Partly sunny
Search
ENG
IN
72%
11:04
09-10-2026


File Edit Selection View Go Run
Experiment8_2
Update
Explorer
Experiment8_2
node_modules
src
components
CourseList.jsx
Navbar.jsx
ProgressBar.jsx
Sidebar.jsx
StatCard.jsx
TaskSection.jsx
App.css
App.jsx
index.css
main.jsx
.gitignore
.oxlintrc.json
index.html
package-lock.json
package.json
vite.config.js
Outline
Timeline
CourseList.jsx
Navbar.jsx
ProgressBar.jsx
Sidebar.jsx
StatCard.jsx
TaskSection.jsx
App.css
App.jsx
src > components > ProgressBar.jsx > default
function ProgressBar({ percent }) {
return (

<div className="progress-fill" style={{ width: `${percent}%` }}>

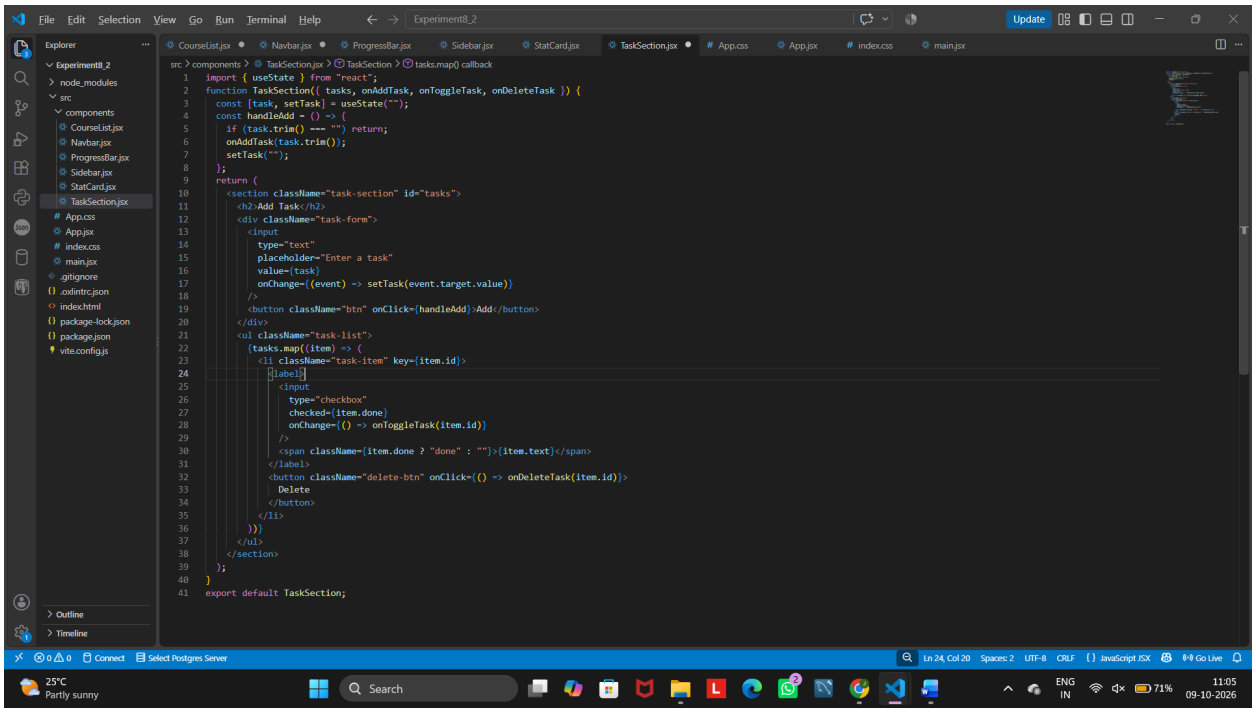
);
}
export default ProgressBar;
0 0
Connect
Select Postgres Server
Ln 9, Col 28
Spaces: 2
UTF-8
CRLF
JavaScript JSX
Go Live
25°C
Partly sunny
Search
ENG
IN
72%
11:04
09-10-2026


File Edit Selection View Go Run
Experiment8_2
Update
Explorer
CourseList.jsx
Navbar.jsx
ProgressBar.jsx
Sidebar.jsx
StatCard.jsx
TaskSection.jsx
App.css
App.jsx
src > components > Sidebar.jsx > default
function Sidebar({ open, onClose }) {
return (
<>
<aside className={open ? "sidebar open" : "sidebar"}>
<a href="#home" onClick={onClose}> Dashboard</a>
<a href="#courses" onClick={onClose}> Courses</a>
<a href="#tasks" onClick={onClose}> Tasks</a>
<a href="#profile" onClick={onClose}> Profile</a>
</aside>
{open && }
</>
);
}
export default Sidebar;
Experiment8_2
node_modules
src
components
CourseList.jsx
Navbar.jsx
ProgressBar.jsx
Sidebar.jsx
StatCard.jsx
TaskSection.jsx
App.css
App.jsx
index.css
main.jsx
.gitignore
.oxlintrc.json
index.html
package-lock.json
package.json
vite.config.js
Outline
Timeline
Connect
Select Postgres Server
Ln 16, Col 24
Spaces: 2
UTF-8
CRLF
JavaScript JSX
Go Live
25°C
Partly sunny
Search
ENG
IN
72%
11:04
09-10-2026

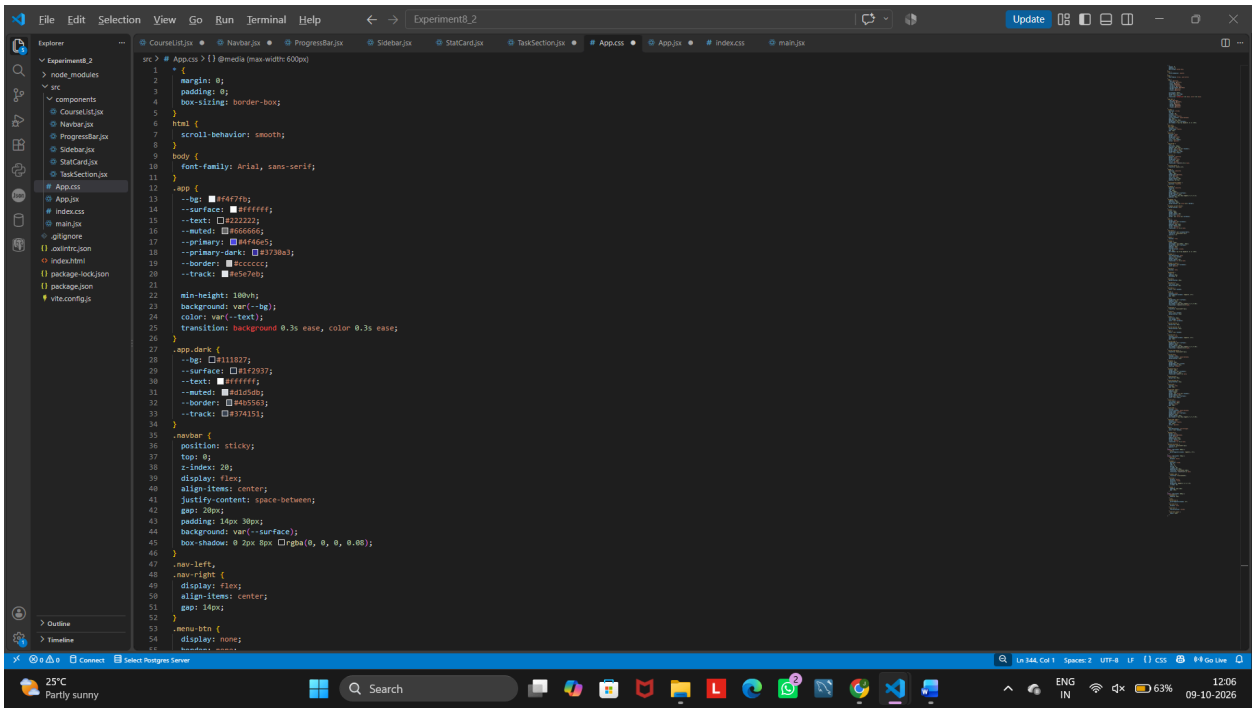

File Edit Selection View Go Run
Experiment8_2
Update
Explorer
CourseList.jsx
Navbar.jsx
ProgressBar.jsx
Sidebar.jsx
StatCard.jsx
TaskSection.jsx
App.css
App.jsx
src > components > StatCard.jsx > default
function StatCard({ title, value }) {
return (

<h3>{title}</h3>
{value}

);
}
export default StatCard;
Experiment8_2
node_modules
src
components
CourseList.jsx
Navbar.jsx
ProgressBar.jsx
Sidebar.jsx
StatCard.jsx
TaskSection.jsx
App.css
App.jsx
index.css
main.jsx
.gitignore
.oxlintrc.json
index.html
package-lock.json
package.json
vite.config.js
Outline
Timeline
Connect
Select Postgres Server
Ln 10, Col 25
Spaces: 2
UTF-8
CRLF
JavaScript JSX
Go Live
25°C
Partly sunny
Search
ENG
IN
72%
11:05
09-10-2026

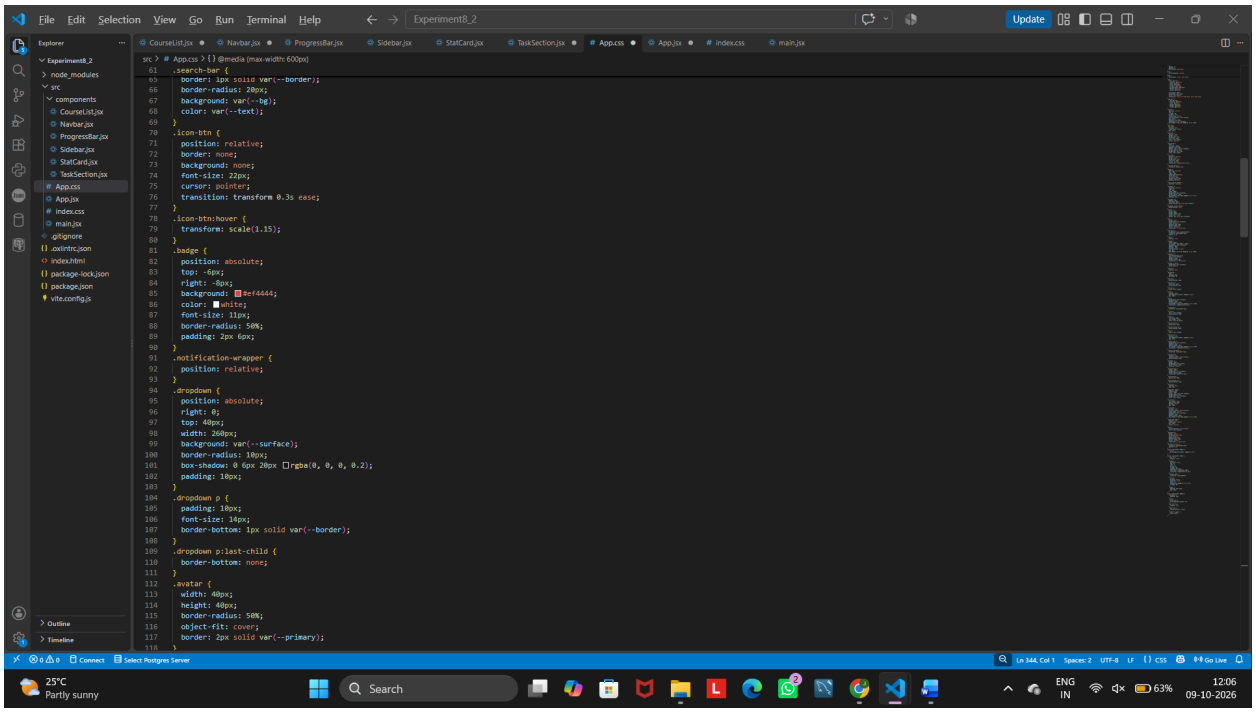


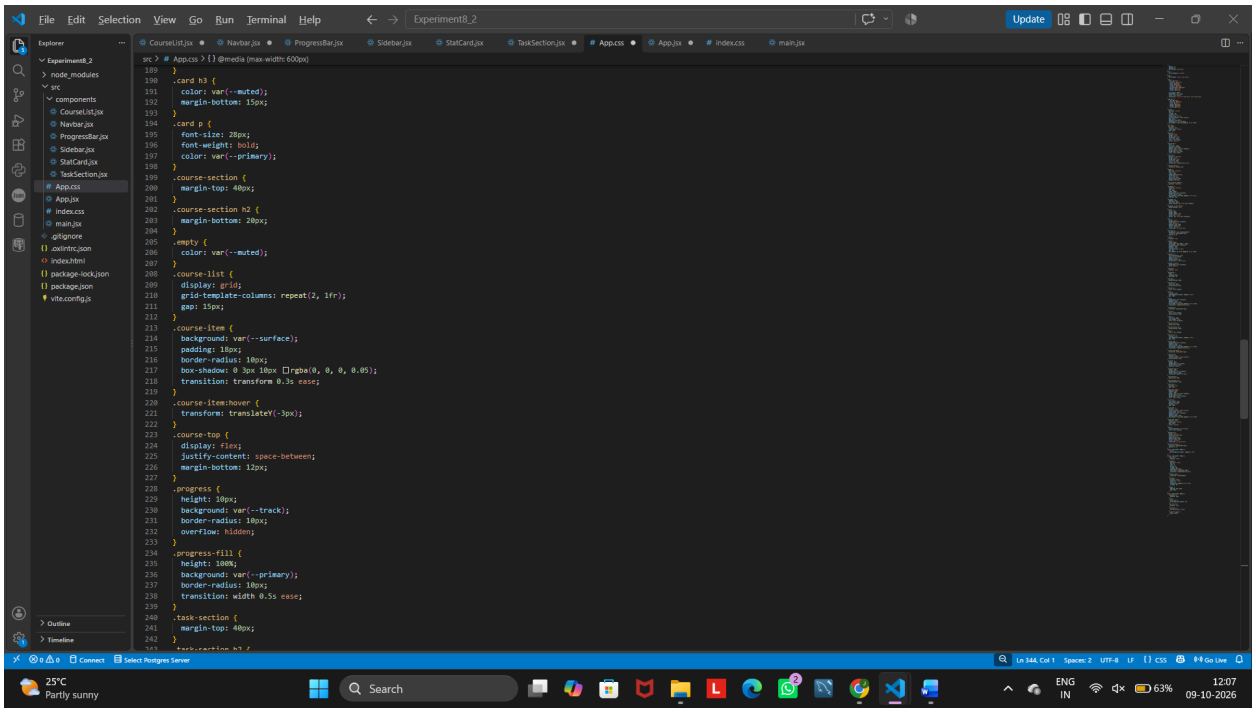
```jsx
import { useState } from "react";
function TaskSection({ tasks, onAddTask, onToggleTask, onDeleteTask }) {
  const [task, setTask] = useState("");
  const handleAdd = () => {
    if (task.trim() === "") return;
    onAddTask(task.trim());
    setTask("");
  };
  return (
    <section className="task-section" id="tasks">
      <h2>Add Task</h2>
      <div className="task-form">
        <input
          type="text"
          placeholder="Enter a task"
          value={task}
          onChange={(event) => setTask(event.target.value)}
        />
        <button className="btn" onClick={handleAdd}>Add</button>
      </div>
      <ul className="task-list">
        {tasks.map((item) => (
          <li className="task-item" key={item.id}>
            <label>
              <input
                type="checkbox"
                checked={item.done}
                onChange={() => onToggleTask(item.id)}
              />
              <span className={item.done ? "done" : ""}>{item.text}</span>
            </label>
            <button className="delete-btn" onClick={() => onDeleteTask(item.id)}>
              Delete
            </button>
          </li>
        ))}
      </ul>
    </section>
  );
}
export default TaskSection;
```

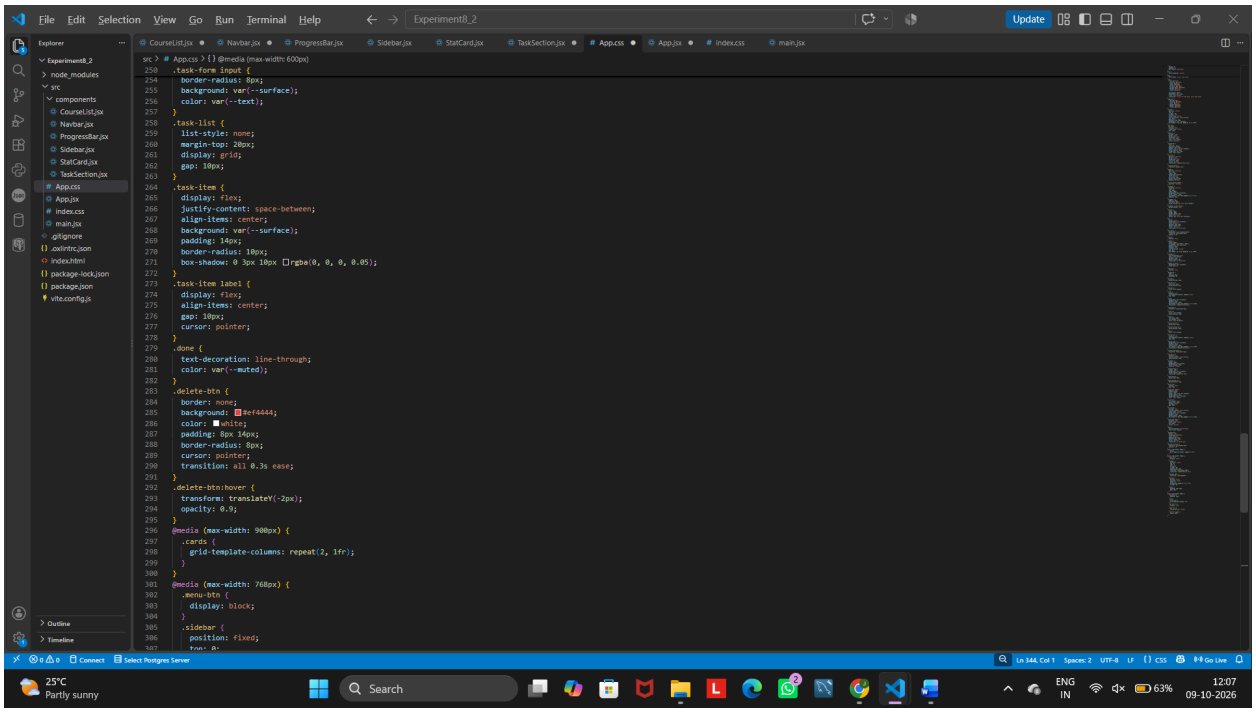


```css
* {
  margin: 0;
  padding: 0;
  box-sizing: border-box;
}
html {
  scroll-behavior: smooth;
}
body {
  font-family: Arial, sans-serif;
}
.app {
  --bg: #f4f7fb;
  --surface: #ffffff;
  --text: #222222;
  --muted: #666666;
  --primary: #4f46e5;
  --primary-dark: #3730a3;
  --border: #cccccc;
  --track: #e5e7eb;

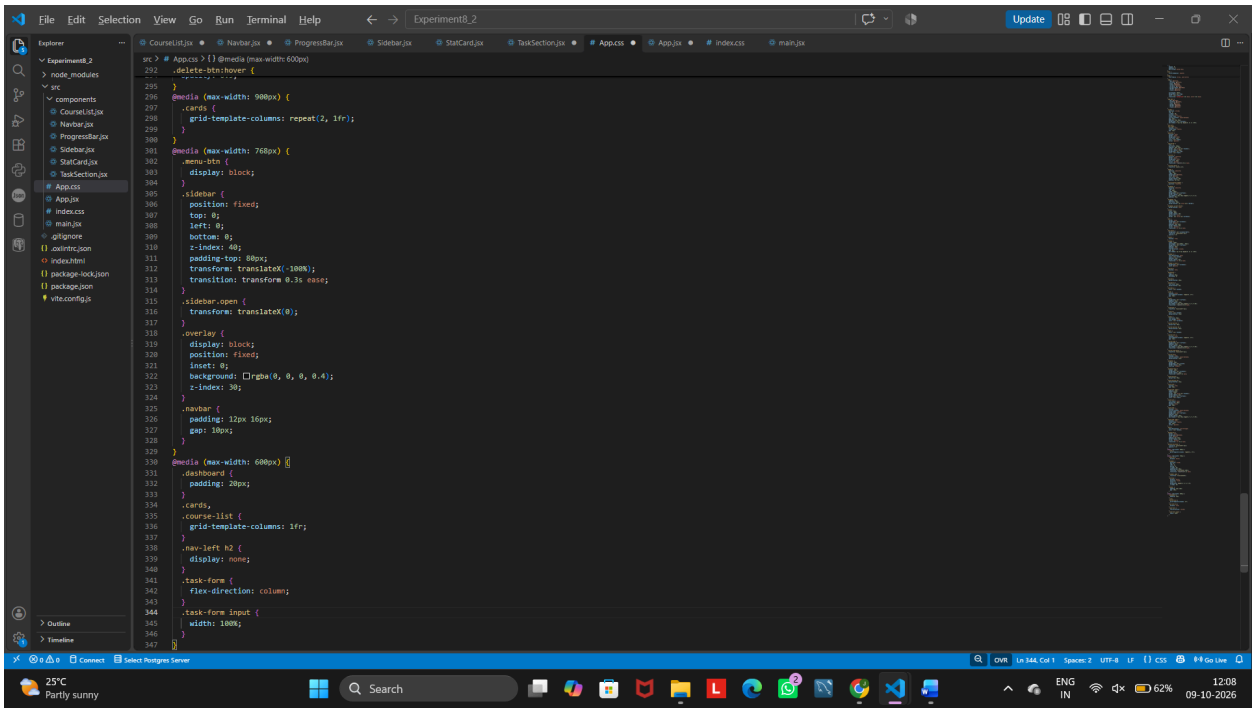
  min-height: 100vh;
  background: var(--bg);
  color: var(--text);
  transition: background 0.3s ease, color 0.3s ease;
}
.app.dark {
  --bg: #111827;
  --surface: #1f2937;
  --text: #ffffff;
  --muted: #d1d5db;
  --border: #4b5563;
  --track: #374151;
}
.navbar {
  position: sticky;
  top: 0;
  z-index: 20;
  display: flex;
  align-items: center;
  justify-content: space-between;
  gap: 20px;
  padding: 14px 30px;
  background: var(--surface);
  box-shadow: 0 2px 8px rgba(0, 0, 0, 0.08);
}
.nav-left,
.nav-right {
  display: flex;
  align-items: center;
  gap: 14px;
}
.menu-btn {
  display: none;
```


File Edit Selection View Go Run Terminal Help
Update
.search-bar {
border: 1px solid var(--border);
border-radius: 20px;
background: var(--bg);
color: var(--text);
}
.icon-btn {
position: relative;
border: none;
background: none;
font-size: 22px;
cursor: pointer;
transition: transform 0.3s ease;
}
.icon-btn:hover {
transform: scale(1.15);
}
.badge {
position: absolute;
top: -6px;
right: -8px;
background: #ef4444;
color: white;
font-size: 11px;
border-radius: 50%;
padding: 2px 6px;
}
.notification-wrapper {
position: relative;
}
.dropdown {
position: absolute;
right: 0;
top: 40px;
width: 260px;
background: var(--surface);
border-radius: 10px;
box-shadow: 0 6px 20px rgba(0, 0, 0, 0.2);
padding: 10px;
}
.dropdown p {
padding: 10px;
font-size: 14px;
border-bottom: 1px solid var(--border);
}
.dropdown p:last-child {
border-bottom: none;
}
.avatar {
width: 40px;
height: 40px;
border-radius: 50%;
object-fit: cover;
border: 2px solid var(--primary);
}
25°C
Partly sunny
Search
ENG
IN
63%
12:06
09-10-2026

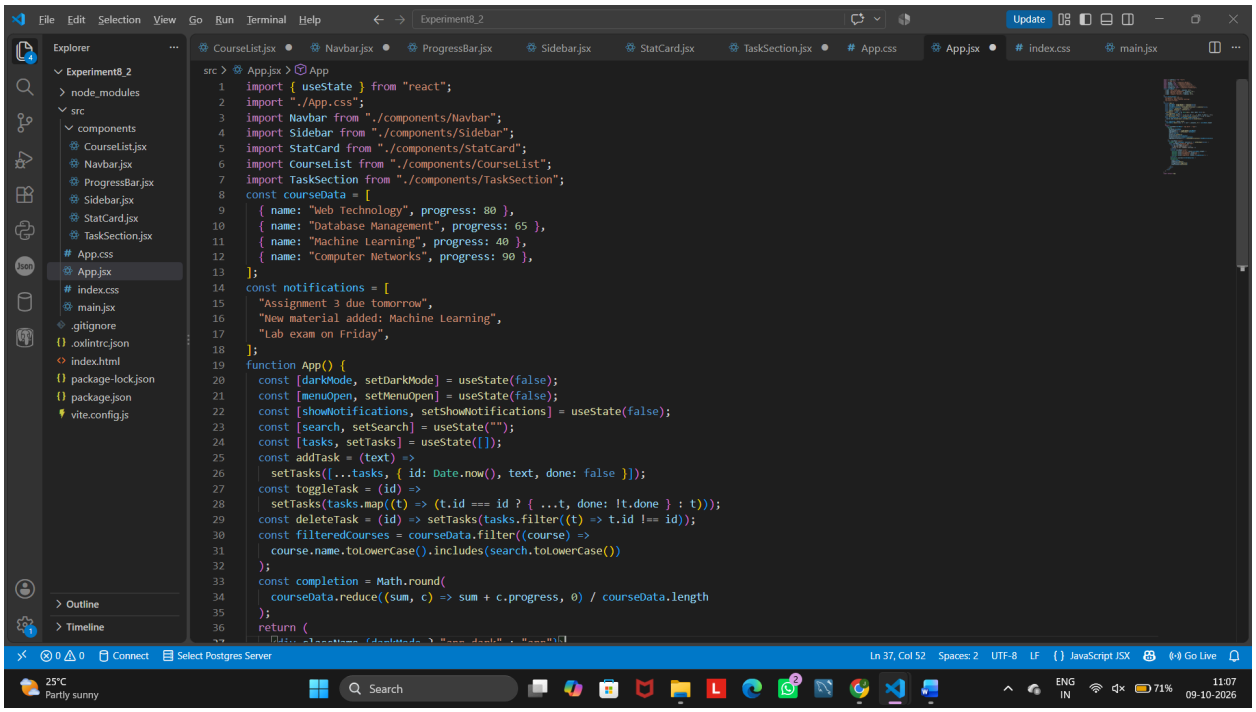

File Edit Selection View Go Run Terminal Help
Update
}
.card h3 {
color: var(--muted);
margin-bottom: 15px;
}
.card p {
font-size: 28px;
font-weight: bold;
color: var(--primary);
}
.course-section {
margin-top: 40px;
}
.course-section h2 {
margin-bottom: 20px;
}
.empty {
color: var(--muted);
}
.course-list {
display: grid;
grid-template-columns: repeat(2, 1fr);
gap: 15px;
}
.course-item {
background: var(--surface);
padding: 18px;
border-radius: 10px;
box-shadow: 0 3px 10px rgba(0, 0, 0, 0.05);
transition: transform 0.3s ease;
}
.course-item:hover {
transform: translateY(-3px);
}
.course-top {
display: flex;
justify-content: space-between;
margin-bottom: 12px;
}
.progress {
height: 10px;
background: var(--track);
border-radius: 10px;
overflow: hidden;
}
.progress-fill {
height: 100%;
background: var(--primary);
border-radius: 10px;
transition: width 0.5s ease;
}
.task-section {
margin-top: 40px;
}
25°C
Partly sunny
Search
ENG
IN
63%
12:07
09-10-2026

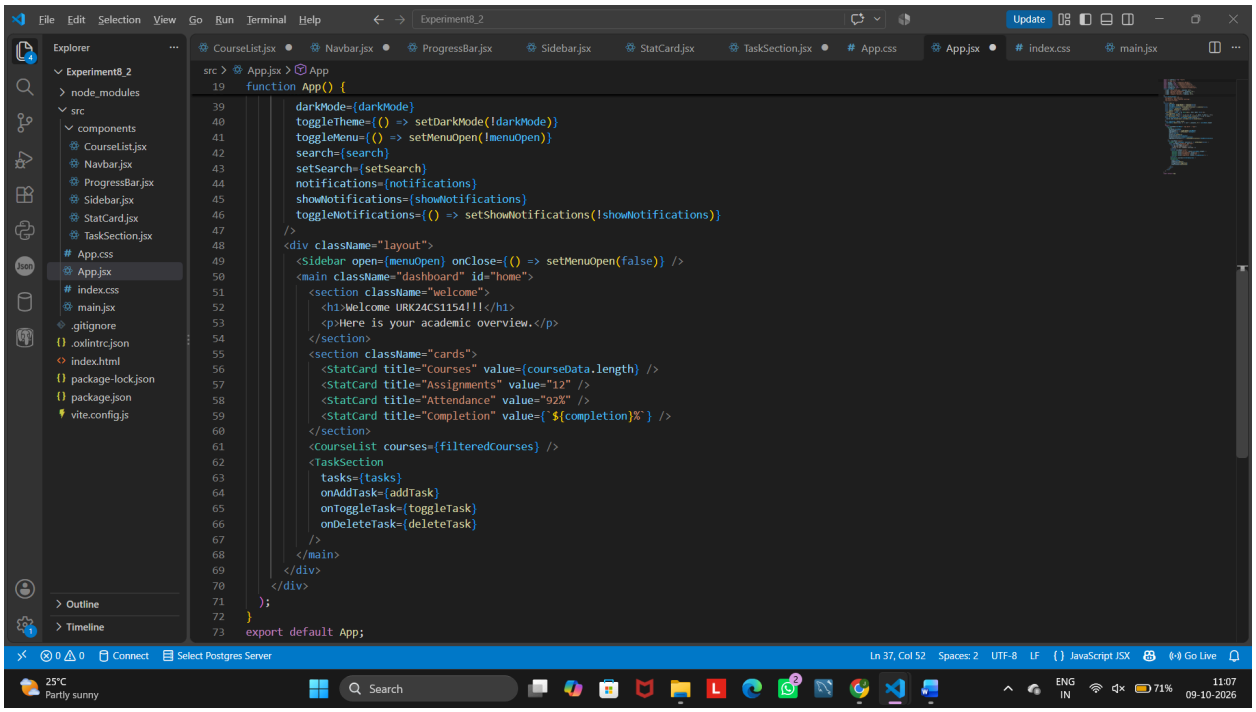


```
.task-form input {
  border-radius: 8px;
  background: var(--surface);
  color: var(--text);
}
.task-list {
  list-style: none;
  margin-top: 20px;
  display: grid;
  gap: 10px;
}
.task-item {
  display: flex;
  justify-content: space-between;
  align-items: center;
  background: var(--surface);
  padding: 14px;
  border-radius: 10px;
  box-shadow: 0 3px 10px rgba(0, 0, 0, 0.05);
}
.task-item label {
  display: flex;
  align-items: center;
  gap: 10px;
  cursor: pointer;
}
.done {
  text-decoration: line-through;
  color: var(--muted);
}
.delete-btn {
  border: none;
  background: #ef4444;
  color: white;
  padding: 8px 14px;
  border-radius: 8px;
  cursor: pointer;
  transition: all 0.3s ease;
}
.delete-btn:hover {
  transform: translateY(-2px);
  opacity: 0.9;
}
@media (max-width: 900px) {
  .cards {
    grid-template-columns: repeat(2, 1fr);
  }
}
@media (max-width: 768px) {
  .menu-btn {
    display: block;
  }
  .sidebar {
    position: fixed;
```



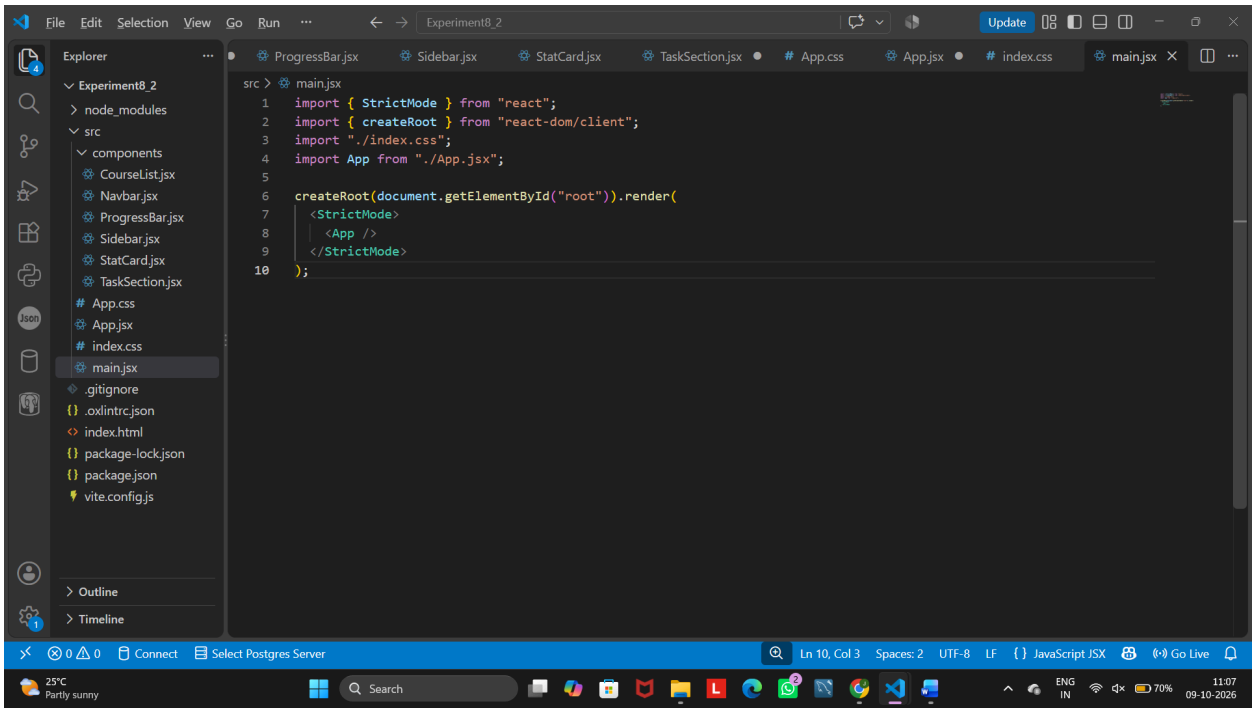
```
.delete-btn:hover {
}
@media (max-width: 900px) {
  .cards {
    grid-template-columns: repeat(2, 1fr);
  }
}
@media (max-width: 768px) {
  .menu-btn {
    display: block;
  }
  .sidebar {
    position: fixed;
    top: 0;
    left: 0;
    bottom: 0;
    z-index: 40;
    padding-top: 80px;
    transform: translateX(-100%);
    transition: transform 0.3s ease;
  }
  .sidebar.open {
    transform: translateX(0);
  }
  .overlay {
    display: block;
    position: fixed;
    inset: 0;
    background: rgba(0, 0, 0, 0.4);
    z-index: 30;
  }
  .navbar {
    padding: 12px 16px;
    gap: 10px;
  }
}
@media (max-width: 600px) {
  .dashboard {
    padding: 20px;
  }
  .cards,
  .course-list {
    grid-template-columns: 1fr;
  }
  .nav-left h2 {
    display: none;
  }
  .task-form {
    flex-direction: column;
  }
  .task-form input {
    width: 100%;
  }
}
```


File Edit Selection View Go Run Terminal Help
Experiment8_2
Update
Explorer
Experiment8_2
node_modules
src
components
CourseList.jsx
Navbar.jsx
ProgressBar.jsx
Sidebar.jsx
StatCard.jsx
TaskSection.jsx
App.css
App.jsx
index.css
main.jsx
.gitignore
.oxlintrc.json
index.html
package-lock.json
package.json
vite.config.js
Outline
Timeline
CourseList.jsx
Navbar.jsx
ProgressBar.jsx
Sidebar.jsx
StatCard.jsx
TaskSection.jsx
App.css
App.jsx
index.css
main.jsx
src > App.jsx > App
import { useState } from "react";
import "./App.css";
import Navbar from "./components/Navbar";
import Sidebar from "./components/Sidebar";
import StatCard from "./components/StatCard";
import CourseList from "./components/CourseList";
import TaskSection from "./components/TaskSection";
const courseData = [
{ name: "Web Technology", progress: 80 },
{ name: "Database Management", progress: 65 },
{ name: "Machine Learning", progress: 40 },
{ name: "Computer Networks", progress: 90 },
];
const notifications = [
"Assignment 3 due tomorrow",
"New material added: Machine Learning",
"Lab exam on Friday",
];
function App() {
const [darkMode, setDarkMode] = useState(false);
const [menuOpen, setMenuOpen] = useState(false);
const [showNotifications, setShowNotifications] = useState(false);
const [search, setSearch] = useState("");
const [tasks, setTasks] = useState([]);
const addTask = (text) =>
setTasks([...tasks, { id: Date.now(), text, done: false }]);
const toggleTask = (id) =>
setTasks(tasks.map((t) => (t.id === id ? { ...t, done: !t.done } : t)));
const deleteTask = (id) => setTasks(tasks.filter((t) => t.id !== id));
const filteredCourses = courseData.filter((course) =>
course.name.toLowerCase().includes(search.toLowerCase())
);
const completion = Math.round(
courseData.reduce((sum, c) => sum + c.progress, 0) / courseData.length
);
return (
Connect
Select Postgres Server
Ln 37, Col 52
Spaces: 2
UTF-8
LF
{ } JavaScript JSX
Go Live
25°C
Partly sunny
Search
ENG
IN
71%
11:07
09-10-2026


File Edit Selection View Go Run Terminal Help
Experiment8_2
Update
Explorer
Experiment8_2
node_modules
src
components
CourseList.jsx
Navbar.jsx
ProgressBar.jsx
Sidebar.jsx
StatCard.jsx
TaskSection.jsx
App.css
App.jsx
index.css
main.jsx
.gitignore
.oxlintrc.json
index.html
package-lock.json
package.json
vite.config.js
Outline
Timeline
function App() {
darkMode={darkMode}
toggleTheme={() => setDarkMode(!darkMode)}
toggleMenu={() => setMenuOpen(!menuOpen)}
search={search}
setSearch={setSearch}
notifications={notifications}
showNotifications={showNotifications}
toggleNotifications={() => setShowNotifications(!showNotifications)}

<Sidebar open={menuOpen} onClose={() => setMenuOpen(false)} />
<main className="dashboard" id="home">
<section className="welcome">
<h1>Welcome URK24CS1154!!!</h1>
Here is your academic overview.
</section>
<section className="cards">
<StatCard title="Courses" value={courseData.length} />
<StatCard title="Assignments" value="12" />
<StatCard title="Attendance" value="92%" />
<StatCard title="Completion" value={`${completion}%`} />
</section>
<CourseList courses={filteredCourses} />
<TaskSection
tasks={tasks}
onAddTask={addTask}
onToggleTask={toggleTask}
onDeleteTask={deleteTask}
/>
</main>

);
}
export default App;
Ln 37, Col 52 Spaces: 2 UTF-8 LF JavaScript JSX Go Live
25°C Partly sunny
Search
ENG IN
71%
11:07 09-10-2026


File Edit Selection View Go Run
Experiment8_2
Update
Explorer
Experiment8_2
node_modules
src
components
CourseList.jsx
Navbar.jsx
ProgressBar.jsx
Sidebar.jsx
StatCard.jsx
TaskSection.jsx
App.css
App.jsx
index.css
main.jsx
.gitignore
.oxlintrc.json
index.html
package-lock.json
package.json
vite.config.js
Outline
Timeline
main.jsx
import { StrictMode } from "react";
import { createRoot } from "react-dom/client";
import "./index.css";
import App from "./App.jsx";
createRoot(document.getElementById("root")).render(
<StrictMode>
<App />
</StrictMode>
);
Ln 10, Col 3 Spaces: 2 UTF-8 LF JavaScript JSX Go Live
25°C Partly sunny
Search
ENG IN
70%
11:07 09-10-2026

## Output

## 25. Concepts Covered

| Concept | Example |
|---|---|
| JSX | <h1>Hello</h1> |
| Components | StatCard |
| Props | titlevalue |
| State | useState() |
| Events | onClick |
| Input handling | onChange |
| Lists | map() |
| Conditional styling | Dark mode |
| CSS Grid | Dashboard cards |
| Media Queries | Responsive UI |

## 26. Viva Questions

1. What is ReactJS?

A JavaScript library for building user interfaces using reusable components.

2. What is JSX?

A syntax that lets you write HTML-like code inside JavaScript.

3. What is a React component?

A reusable piece of UI written as a function that returns JSX.

4. Functional vs class components?

Functional components are simple functions that use hooks, while class components use ES6 classes and lifecycle methods.

5. What are props?

Read-only data passed from a parent component to a child component.

6. What is state?

Data inside a component that can change and makes the UI update when it does.

7. What is useState()?

A hook that adds a state variable and an update function to a functional component.

8. Props vs state?

Props are passed in from outside and cannot be changed by the child, while state is owned by the component and can be changed.

9. What is event handling in React?

Responding to user actions like clicks or typing using handlers such as onClick and onChange.

10. What is a controlled component?

An input whose value is controlled by React state through value and onChange.

11. Why is the key attribute used in lists?

It gives each list item a unique identity so React can update the list efficiently.

12. What is responsive web design?

Designing a page so it adapts to different screen sizes like mobile, tablet and desktop.

13. What is CSS Flexbox?

A one-dimensional layout system that arranges items in a row or column.

14. What is CSS Grid?

A two-dimensional layout system that arranges items in rows and columns.

15. What is the purpose of media queries?

To apply different CSS styles depending on screen width or device type.

16. What is component reusability?

Writing a component once and using it many times with different props.

17. Why divide large React apps into components?

It makes the code easier to read, reuse, test and maintain.

18. What is the purpose of npm?

It installs and manages the packages and dependencies a project needs.

19. What is Vite?

A fast build tool and development server for creating modern React projects.

20. What happens when React state changes?

React re-renders the component and updates the screen with the new data.

## Result

Thus, a modern and responsive user interface was successfully designed and implemented using ReactJS with reusable components, JSX, props, state, event handling, CSS styling, responsive layouts, and theme switching.